Date: 14 MAY 2026 **Regd. No.** | | | | | | | | | | |

Sub Code:23AM5T01 **MIC23**

B. Tech V Semester Supplementary Examinations, April-2026

DEEP LEARNING

(AI&ML)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	What is principal component analysis ?	2 M	L1	CO1
	b)	Explain the concept of linear separability.	2 M	L1	CO1
	c)	Define empirical risk minimization.	2 M	L1	CO2
	d)	What is regularization in deep neural networks?	2 M	L1	CO2
	e)	Explain Adam optimizer.	2 M	L2	CO3
	f)	What is batch normalization?	2 M	L1	CO3
	g)	Differentiate between GRU and LSTM.	2 M	L4	CO4
	h)	What are Restricted Boltzmann Machines?	2 M	L1	CO4
	i)	Define Agentic AI with one example.	2 M	L1	CO5
	j)	What for GPT models used?	2 M	L1	CO5
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Explain the concept of eigenvalues and eigenvectors with an example.	5M	L2	CO1
	b)	Describe the Perceptron Learning Algorithm and its convergence theorem.	5M	L1	CO1
		(OR)			
2.	**a)**	Describe threshold logic using perceptrons.	5M	L1	CO1
	b)	Write short notes on biological neuron and computational units.	5M	L1	CO1
		UNIT – II			
3.	**a)**	Explain the challenges in training deep neural networks.	5M	L3	CO2
	b)	Explain backpropagation with a suitable example.	5M	L2	CO2
		(OR)			
4.	**a)**	What is empirical risk minimization? Illustrate with an example.	5M	L1	CO2
	b)	Derive the weight update rule using gradient descent.	5M	L2	CO2
		UNIT – III			
5.	**a)**	Explain the working of Adadelta optimizer.	5M	L2	CO3
	b)	Discuss the role of Nesterov Accelerated Gradient (NAG).	5M	L6	CO3
		(OR)			

6.	**a)**	What is the saddle point problem? How can it be addressed?	5M	L1	CO3
	b)	Compare Adagrad and RMSProp optimization techniques.	5M	L4	CO3
		UNIT – IV			
7.	**a)**	Explain convolution operation with types of convolutions.	5M	L2	CO4
	b)	Discuss bidirectional RNNs and their applications.	5M	L6	CO4
		(OR)			
8.	**a)**	Write short notes on LeNet and AlexNet architectures.	5M	L1	CO4
	b)	What is Gibbs sampling? Explain its role in RBMs.	5M	L1	CO4
		UNIT – V			
9.	**a)**	What are Transformers? Explain with architecture diagram.	5M	L1	CO5
	b)	Explain the architecture of a VariationalAutoencoder.	5M	L1	CO5
		(OR)			
10.	**a)**	What are the types of Generative AI? Give suitable examples.	5M	L1	CO5
	b)	Discuss real-world applications of Transformers in NLP.	5M	L6	CO5

***** End of the Question Paper *****

Date: 19 MAY 2026 **Regd. No.** | | | | | | | | | | |

Sub Code:23AM5T03 **MIC23**

B. Tech V Semester Supplementary Examinations, April-2026

OPERATING SYSTEMS

(AI&ML)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	Define batch processing system.	2 M	L1	CO1
	b)	What are layered operating systems?	2 M	L1	CO1
	c)	What are the different states of a process?	2 M	L1	CO2
	d)	Differentiate preemptive and non-preemptive scheduling.	2 M	L2	CO2
	e)	Define busy waiting.	2 M	L1	CO3
	f)	What is starvation?	2 M	L1	CO3
	g)	Define dynamic linking.	2 M	L1	CO4
	h)	What is a page fault?	2 M	L1	CO4
	i)	List the attributes of a file.	2 M	L1	CO5
	j)	Define free space management.	2 M	L1	CO5
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Explain in detail the functions of an operating system.	5M	L2	CO1
	b)	Describe the types of system calls with examples.	5M	L2	CO1
		(OR)			
2.	**a)**	Discuss the importance of debugging in operating system development.	5M	L2	CO1
	b)	Explain in detail the structure of an operating system	5M	L2	CO1
		UNIT – II			
3.	**a)**	Given the processes below, schedule them using FCFS. Process | Arrival time | Burst time P1 | 1 | 4 P2 | 3 | 5 P3 | 5 | 2 P4 | 7 | 3 Calculate: (i) Completion Time for each process (ii) Average Waiting Time (iii) Average Turnaround Time	5M	L3	CO2
	b)	Describe message-passing and shared-memory models of IPC.	5M	L2	CO2

		(OR)			
4.	**a)**	Discuss multithreading models in detail	5M	L2	CO2
	b)	Discuss inter-process communication (IPC) and its mechanisms.	5M	L2	CO2
		UNIT – III			
5.	**a)**	Explain how Dining Philosophers Problem can be solved using semaphores.	5M	L2	CO3
	b)	Explain deadlock avoidance with an example of Banker's Algorithm.	5M	L2	CO3
		(OR)			
6.	**a)**	Explain Readers–Writers Problem using semaphores.	5M	L2	CO3
	b)	Discuss the necessary conditions for deadlock with examples.	5M	L2	CO3
		UNIT – IV			
7.	**a)**	Discuss demand paging and describe page fault handling.	5M	L2	CO4
	b)	Explain contiguous memory allocation and discuss fragmentation issues.	5M	L2	CO4
		(OR)			
8.	**a)**	What is Translation Lookaside Buffer (TLB)? Explain its working with a diagram.	5M	L2	CO4
	b)	Explain FCFS and SSTF disk scheduling algorithms with examples.	5M	L2	CO4
		UNIT – V			
9.	**a)**	Discuss different access methods of a file system with examples.	5M	L2	CO5
	b)	Explain file system mounting and its significance.	5M	L2	CO5
		(OR)			
10.	**a)**	Discuss file sharing mechanisms and protection issues.	5M	L2	CO5
	b)	Explain file allocation methods.	5M	L2	CO5

***** End of the Question Paper *****

Date: 16 MAY 2026 **Regd. No.** | | | | | | | | | | |

Sub Code:23CS/AM5T02 **MIC23**

B. Tech V Semester Supplementary Examinations, April-2026

COMPUTER NETWORKS

(Common to CSE, AI&ML)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	Mention two key differences between OSI and TCP/IP models.	2 M	L1	CO1
	b)	What is coaxial cable and where is it commonly used?	2 M	L1	CO1
	c)	Define flow control?	2 M	L1	CO2
	d)	What is one's complement checksum? Where is it used?	2 M	L1	CO2
	e)	What is random access in MAC protocols?	2 M	L1	CO3
	f)	Define Carrier Sense Multiple Access.	2 M	L1	CO3
	g)	Define a subnet.	2 M	L1	CO4
	h)	List the services provided by the network layer to the transport layer.	2 M	L1	CO4
	i)	What is the role of the transport layer in the OSI model?	2 M	L1	CO5
	j)	Differentiate between local and remote logging.	2 M	L1	CO5
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Explain the OSI reference model.	5M	L2	CO1
	b)	Explain the following network topologies with diagrams, advantages, and disadvantages: Ring topology and Mesh topology.	5M	L2	CO2
		(OR)			
2.	**a)**	Explain in detail about fiber optic cable.	5M	L2	CO1
	b)	Explain the differences between LAN, MAN, and WAN.	5M	L2	CO1
		UNIT – II			
3.	**a)**	Explain fixed-size framing with an example.	5M	L2	CO2
	b)	Explain Checksum with suitable example.	5M	L2	CO2
		(OR)			
4.	**a)**	Explain one-bit sliding window protocol with diagram and example.	5M	L2	CO2
	b)	Explain Point-to-Point Protocol.	5M	L2	CO2
		UNIT – III			
5.	**a)**	Explain reservation-based MAC with an example.	5M	L2	CO3
	b)	Explain FDMA and TDMA.	5M	L2	CO3
		(OR)			
6.	**a)**	Explain Standard Ethernet and its characteristics.	5M	L2	CO3
	b)	Explain working of token passing with example.	5M	L2	CO3
		UNIT – IV			
7.	**a)**	Explain the design issues in the network layer.	5M	L2	CO4
	b)	Explain the Distance vector routing algorithms with example.	5M	L2	CO4

		(OR)			
8.	**a)**	Explain the IPv6 header format in detail.	5M	L2	CO4
	b)	Explain methods to connect networks.	5M	L2	CO4
		UNIT – V			
9.	**a)**	List five differences between UDP and TCP.	5M	L2	CO5
	b)	Explain the Transmission Control Protocol.	5M	L2	CO5
		(OR)			
10.	**a)**	Explain HTTP protocol in detail.	5M	L2	CO5
	b)	Explain the Domain Name System.	5M	L2	CO5

***** End of the Question Paper *****

Date: 21 MAY 2026 **Regd. No.** | | | | | | | | | | |

Sub Code: 23MB5T01 **MIC23**

B. Tech V Semester Supplementary Examinations, April-2026

ENTREPRENEURSHIP DEVELOPMENT AND VENTURE CREATION

(Common to CSE, AI&DS, AI&ML)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	List any two skillsets essential for entrepreneurship.	2 M	L1	CO1
	b)	Define intrapreneurial leadership.	2 M	L1	CO1
	c)	What is meant by customer segmentation?	2 M	L1	CO2
	d)	Define problem-customer fit.	2 M	L1	CO2
	e)	What is proof-of-concept testing?	2 M	L1	CO3
	f)	Define prototype and Minimum Viable Product (MVP).	2 M	L1	CO4
	g)	What are fixed and variable costs?	2 M	L1	CO5
	h)	State any two sources of funding for start-ups.	2 M	L1	CO5
	i)	List any two key components of persuasive storytelling.	2 M	L1	CO6
	j)	Mention any two elements that attract investors in a start-up pitch.	2 M	L1	CO6
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	Illustrate the mindset required for successful entrepreneurial leadership.	5M	L2	CO1
	b)	Explain how role models influence entrepreneurial behavior and development.	5M	L2	CO1
		(OR)			
2.	**a)**	Associate how entrepreneurship contributes to national economic growth.	5M	L2	CO1
	b)	Discuss how students can develop entrepreneurial skillsets while on campus.	5M	L2	CO1
		UNIT – II			
3.	**a)**	Explain the potential challenges your startup might face due to competition.	5M	L2	CO2
	b)	Interpret the process of identifying and defining a problem using Design Thinking principles.	5M	L2	CO2
		(OR)			

4.	**a)**	Find the direct competitors and alternatives for the idea of a "smart water bottle that tracks hydration,".	5M	L3	CO2
	b)	Analyze how mapping competition and industry trends assists in assessing the feasibility of a startup opportunity.	5M	L4	CO2
UNIT – III					
5.	**a)**	Discuss how iterative prototyping helps in developing an innovative product.	5M	L2	CO3
	b)	Demonstrate how competition analysis supports effective opportunity assessment.	5M	L3	CO3
(OR)					
6.		Assess the challenges faced by entrepreneurs during proof-of-concept testing and suggest strategies to overcome them.	10M	L5	CO4
UNIT – IV					
7.	**a)**	Explain why knowing the value proposition is important for a business.	5M	L2	CO5
	b)	Summarize the key components of a financial plan and their significance in achieving profitability.	5M	L2	CO5
(OR)					
8.	**a)**	Describe the key components of a compelling entrepreneurial story	5M	L2	CO5
	b)	Categorize different forms of business organization, identify sources of funds.	5M	L4	CO5
UNIT – V					
9.	**a)**	Explain the importance of scalability while planning the growth of a venture.	5M	L2	CO6
	b)	Describe the process of refining a business idea to make it investor-ready.	5M	L2	CO6
(OR)					
10.		Examine the key challenges entrepreneurs face while preparing for venture scaling and propose suitable strategies to overcome them.	10M	L5	CO6

***** End of the Question Paper *****